

Securing the Skies: A Comprehensive Survey on Anti-UAV Methods, Benchmarking, and Future Directions

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Abstract

Unmanned Aerial Vehicles (UAVs) are indispensable for infrastructure inspection, surveillance, and related tasks, yet they also introduce critical security challenges. This survey provides a wide-ranging examination of the anti-UAV domain, centering on three core objectives—classification, detection, and tracking—while detailing emerging methodologies such as diffusion-based data synthesis, multi-modal fusion, vision-language modeling, self-supervised learning, and reinforcement learning. We systematically evaluate state-of-the-art solutions across both single-modality and multi-sensor pipelines (spanning RGB, infrared, audio, radar, and RF) and discuss large-scale as well as adversarially oriented benchmarks. Our analysis reveals persistent gaps in real-time performance, stealth detection, and swarm-based scenarios, underscoring pressing needs for robust, adaptive anti-UAV systems. By highlighting open research directions, we aim to foster innovation and guide the development of next-generation defense strategies in an era marked by the extensive use of UAVs.

1. Introduction

Unmanned Aerial Vehicles (UAVs) have expanding roles in civilian and military domains, offering benefits for infrastructure inspection, reconnaissance, and surveillance. Yet their rapid proliferation also introduces serious security threats, including unauthorized entry into restricted airspace and covert data gathering. As low-cost, agile drones become more prevalent, the need for robust counter-UAV (anti-UAV) technologies that can detect, track, and neutralize hostile UAVs grows increasingly urgent.

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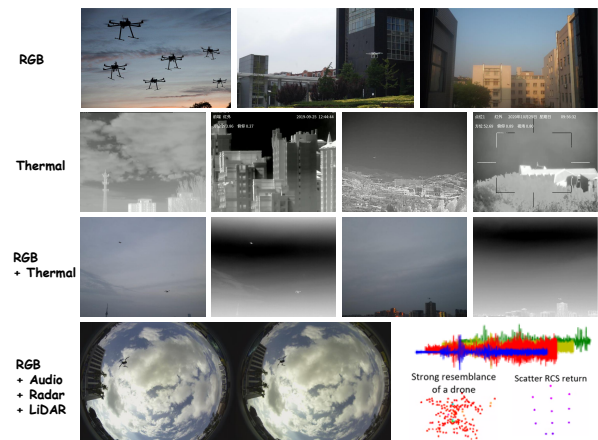


Figure 1. Datasets capturing UAV with diverse modalities: 1). RGB [163], 2). Thermal [51]; 3). RGB + Thermal [153]; 4). RGB + Audio + Radar + LiDAR [156].

Scope & Motivation. This survey provides a comprehensive overview of current anti-UAV research, focusing on three key objectives: *classification* (Section 3), *detection* (Section 4), and *tracking* (Section 5). We underscore the importance of leveraging multi-modal data—encompassing RGB, infrared (IR), radar, RF, and audio—to address small-object detection, erratic flight trajectories, and adversarial tactics. Figure 2 illustrates five primary research thrusts that drive modern anti-UAV innovations: (i) diffusion-based data synthesis and domain adaptation, (ii) vision-language modeling, (iii) self-supervised and unsupervised learning, (iv) multi-modal fusion, and (v) reinforcement learning.

Datasets & Benchmarking. The advancement of anti-UAV systems crucially depends on high-quality, large-scale datasets. Section 2 reviews pivotal benchmarks—including *DroneRF*, *Halmstad Drone*, and *Anti-UAV*—detailing their annotation fidelity, sampling rates, and environmental diversity. We also highlight the pressing need for more specialized datasets to cover scenarios such as night-time op-

Table 1. **Statistical Comparison of Public UAV Datasets.** “Seq.” = number of sequences, “Frame” = number of frames, “T-Type” = target types, “S-Type” = scenario types. “Modality”: RGB, IR, Audio (A), Radar (RA), LiDAR (L), RF. “Task”: Classification (C), Detection (D), Tracking (T). “Annotation”: class, mask, bbox, centroid, 3D coords (3D coordinates), or trajectory.

Dataset	Task	Modality	Seq.	Frame	Annotation	Resolution	Rate (Hz)	T-Type	S-Type	Venue (Year)
FL-Drones [103]	D, T	RGB	14	39K	Bbox	640×480 / 752×480	–	1	2	TPAMI (2016)
NPS-Drones [68]	D, T	RGB	50	70K	Bbox	1920×1280 / 1280×760	–	1	–	IROS (2016)
USC Drone [24]	D, T	RGB	60	20K	Bbox	1920×1080	15	1	–	APSIPA ASC (2017)
DroneRF [8]	C, D	RF	454	10M	Class	14-bit (RF)	Less than 1K (RF)	3	–	Data in Brief (2019)
DroneAudio [2]	C, D	A	–	1332 clips	Class	–	16K (A)	2	–	IWCMC (2019)
MAV-VID [100]	D, T	RGB	–	40K	Bbox	–	–	1	–	IEEE Access (2020)
Hui [55]	D, T	IR	22	16K	Centroid	256×256	100	1	22	CSD (2020)
Acoustic-UAV [21]	C, D	A	–	91K sec	Class	–	22.05K (A)	9	6	Drones (2021)
mDrone [165]	C, D	RA	100	60K	3D coords	–	10 (RA)	1	–	ICRA (2021)
Det-Fly [166]	D	RGB	–	13K	Bbox	3840×2160	5	1	4	RA-L (2021)
MOT-Fly [28]	C, D	RGB	16	11K	Bbox	1920×1080	–	3	–	ICUS (2021)
Halmstad Drone [122]	D, T	RGB, IR, A	650	203K	Bbox	640×512 (RGB), 320×256 (IR)	60 (IR), 30 (RGB)	4	3	ICPR (2021)
Drone-vs-Bird [29]	C, D	RGB	77	105K	Bbox	720×576 – 3840×2160	–	8	–	AVSS (2021)
Anti-UAV [58]	D, T	RGB, IR	318	297K	Bbox	1920×1080 (RGB), 640×512 (IR)	25	–	7	TMM (2021)
UAVSwarm [133]	D, T	RGB	72	12K	Bbox	446×276 – 1919×1079	–	19	13	Remote Sens. (2022)
Synthetic Drone [140]	C	RGB	–	2K	Class	256×256	–	4	10	CORD (2022)
DUT Anti-UAV [163]	D, T	RGB	20	35K	Bbox	160×240 – 3744×5616	–	1	7	T-ITS (2022)
AOT [10]	D, T	IR	4943	5.9M+	Bbox	2448×2048	10	–	–	AWS (2023)
Anti-UAV600 [171]	D, T	IR	600	723K	Bbox	640×512	25	1	–	Preprint (2023)
Anti-UAV410 [51]	D, T	IR	410	438K	Bbox	640×512	25	1	7	TPAMI (2024)
MMAUD [156]	C, D, T	RGB, A, RA, L	50	45K	Trajectory	2560×960 (RGB)	30 (RGB), 41.8 (A), 15 (RA), 10 (L)	6	–	ICRA (2024)
RGBT-Tiny [153]	D, T	RGB, IR	115	93K	Bbox	640×512	15	7	8	TPAMI (2025)

erations, swarm-based threats, and real-time interception. Table 1 contrasts these datasets, revealing persistent gaps in annotation standards, sensor synchronization, and adversarial realism, which limit standardized evaluation and hinder robust system development.

Classification, Detection, & Tracking. While detection locates UAVs in visual or sensor data, classification provides deeper insight into UAV types and flight profiles. In Section 3, we survey recent classification methods that integrate visual and non-visual features for improved accuracy and threat assessment. Section 4 examines state-of-the-art detection pipelines ranging from traditional feature-based to CNN- and transformer-based architectures, whereas Section 5 discusses leading tracking algorithms (e.g., YOLO-based, Siamese) designed to manage fast-moving UAVs in cluttered or adversarial settings.

Emerging Techniques & Future Directions. Section 6 spotlights key emerging paradigms—such as diffusion-driven augmentation, vision-language integration, self-supervised learning, advanced multi-modal fusion, and reinforcement learning for real-time control—and discusses their tremendous potential to reshape anti-UAV capabilities. Ultimately, these methods aim to enable adaptable, real-time systems equipped to handle evolving threats through domain adaptation, robust sensor integration, and rapid policy updates. Section 6 concludes with a forward-looking perspective on next-generation anti-UAV solutions, outlining how these emerging methodologies can drive future research and industrial innovation.

2. UAV Datasets & Benchmarking

Table 1 summarizes publicly available UAV datasets spanning various sensor modalities (e.g., RGB, IR, audio, radar, LiDAR, and RF), annotation schemes (e.g., bounding boxes, 3D coordinates), and sampling rates. This section highlights significant dataset initiatives, explores emerging trends, and underscores unresolved challenges.

Foundational Efforts & Modality Expansion. Initial benchmarks such as *FL-Drones* [103], *NPS-Drones* [68], and *USC Drone* [88, 138] provided core RGB datasets for aerial detection and tracking. Beyond the visual domain, *DroneRF* [8] captures electromagnetic signatures for RF-based UAV classification, while *DroneAudio* [2] and *Acoustic-UAV* [21] focus on acoustic features. More recent efforts extend coverage to multi-modal data: *Halmstad Drone* [120, 121] fuses RGB, IR, and audio, and *MMAUD* [156] integrates radar and LiDAR. Community-driven competitions (e.g., the 2nd and 3rd Anti-UAV Workshops [162, 164] and the 4th Anti-UAV Challenge [89]) feature adversarial or cluttered scenarios that reflect real-world operational complexity, while *UAVSwarm* [133] and *mDrone* [165] emphasize large-scale swarm coordination or low-visibility tracking—both highlighting the need for robust multi-sensor fusion [102].

Complex Scenes & Modalities. Recent datasets target increasingly demanding environments and more granular annotations. For instance, *Anti-UAV600* [171] and *Anti-UAV410* [51] rely on infrared or thermal imaging to capture nighttime or harsh weather conditions, while *Drone-*

Table 2. Representative UAV Classification Approaches by Modality. *Disclaimer:* All entries summarize our own interpretation of each paper’s methodology and reported outcomes. Since authors use distinct datasets, labels, and metrics, **no direct numerical comparison** is valid. *Abbrev.:* UAV = Unmanned Aerial Vehicle, CNN = Convolutional Neural Network, RNN = Recurrent Neural Network, SVM = Support Vector Machine, DNN = Deep Neural Network, GAN = Generative Adversarial Network, LSTM = Long Short-Term Memory, CLSTM = Contextual LSTM, DDI = Drone Detection and Identification, PSD = Power Spectral Density, MFCC = Mel-Frequency Cepstral Coefficients, LFCC = Linear Frequency Cepstral Coefficients, SCF = Spectral Correlation Functions, TWM = Time-frequency Waterfall Map, GTCC = GammaTone Cepstral Coefficients, SNR = Signal-Noise Ratio, APE = Average Pose Error, CV = Cross Validation, IR = Infrared, ID = Identification, Radar = RA, LiDAR = L.

Ref. (Yr.)	Mod.	Method	Dataset	Key Observations (Our Reading)
<i>RF-Based</i>				
Al-Sa’d et al. (2019) [3]	RF	3 DNNs (RF fingerprint)	DroneRF [8]	Introduced DroneRF (open-source, multiple flight modes). Developed three DNN classifiers for UAV presence, type, and flight mode. Achieved 99.7% (2-class), 84.5% (4-class), 46.8% (10-class). Forms an early baseline on DroneRF.
Al-Emadi et al. (2020) [1]	RF	CNN	DroneRF [8]	Achieved 99.8% (2-class), 85.8% (4-class), and 59.2% (10-class). Comparable to Al-Sa’d et al. [3] on 2-class tasks, showing CNN’s viability for RF-based identification.
Medaiyese et al. (2021) [80]	RF	XGBoost pipeline	DroneRF [8]	Used 227 signal segments. Reached 99.96% (2-class), 90.73% (4-class), 70.09% (10-class). Overall stronger than Al-Sa’d et al. [3] for simpler tasks, but lags on higher-class classification vs. more recent solutions.
Li et al. (2021) [72]	RF	Bispectrum Siamese Network + Contrastive Learning	DroneRF [8]	Converts UAV signals to bispectrum for unsupervised representation. Achieved 100% (2-class), 98.57% (4-class), and 92.31% (10-class). Gains up to 10% over SimCLR, SimSiam, and GAN baselines, though still below XGBoost for large class sets.
Nemer et al. (2021) [86]	RF	Ensemble learning with hierarchical classification	DroneRF [8]	Multi-stage scheme yields 99% overall accuracy on ten-type classification. Outperforms earlier CNN approaches (e.g., [1]) for flight-mode tasks.
Inani et al. (2023) [56]	RF	XGBoost + PSD features + IDCNN	DroneRF [8]	Achieved 100% (2-class), 99.82% (4-class), 99.51% (10-class)—current state-of-the-art on DroneRF. Improves upon [80] and [3] across all class splits, leveraging PSD feature fusion with deep learning.
<i>Audio-Based</i>				
Al-Emadi et al. (2019) [2]	Aud.	CNN, RNN, CRNN	DroneAudio [2]	Released DroneAudio dataset with ~ 1300 drone clips. Demonstrated CNN-based acoustic ID, achieving 96.38% (2-class) and 92.94% (3-class) accuracy. Provided a baseline for audio-based UAV detection.
Katta et al. (2022) [61]	Aud.	CRNN, CNN, LSTM, CLSTM, Transformer	DroneAudio [2]	Benchmarked multiple neural architectures for UAV detection and ID on DroneAudio. LSTM outperformed CRNN/CNN with 98.93% accuracy (2-class) and 98.60% accuracy (3-class).
Xiao et al. (2025) [148]	Aud.	CNN+RNN (temporal)	MMAUD (audio-only) [156]	Achieved 98.0% classification accuracy and 0.55 APE for 3D trajectory estimation with a Mamba-based audio pipeline (4-channel short melspectrograms). Demonstrated robust performance across day/night conditions.
<i>Radar-Based</i>				
Mendis et al. (2016) [81]	Radar	CNN (micro-Dopp.)	None public	Utilized deep belief networks for micro-Doppler SCF signatures, identifying UAV presence within radar beam width. One of the earliest radar-based UAV classification approaches, but lacks open data for direct replication.
<i>Vision-Based</i>				
Wisniewski et al. (2022) [139]	RGB	DenseNet201 CNN	Synthetic Drone [140]	Achieved 92.4% classification accuracy (distinguishing DJI Phantom, Mavic, and Inspire) with 88.8% precision, 88.6% recall, 88.7% F1-score. Demonstrated synthetic data viability for UAV type classification.
<i>Multi-Modal</i>				
Deng et al. (2024) [32]	RGB, RA, L	Deep fusion	MMAUD [156]	Explored 4-class UAV classification in 3D pose estimation. Fused LiDAR, radar, and fisheye camera inputs via an EfficientNet-B7-based pipeline. Achieved 81.36% classification accuracy, improving keyframe-based multi-sensor approaches.
Xiao et al. (2024) [147]	RGB, Aud.	AV-DTEC (self-supervised audio-visual fusion)	MMAUD [156]	Achieved 99.3% classification accuracy and 0.67m APE. Parallel state-space fusion boosted performance under varied illumination. Offers an alternative to full multi-sensor pipelines by only focusing on audio-visual synergy.

vs-Bird [29] focuses on distinguishing UAVs from visually similar wildlife. Other repositories (e.g., *Midgard* [131] and *Rosner* [102]) explore indoor or GPS-denied settings, alongside multi-view or re-identification tasks [11, 59]. Simultaneously, annotations have expanded from bounding boxes to 3D coordinates and comprehensive trajectory information [156], aligning with the growing complexity of interception and multi-UAV maneuvers.

Trends & Open Challenges. Despite notable advances in dataset scope, several gaps persist. Collections like *MMAUD* [156] and *Halmstad Drone* [121] require precise sensor calibration and synchronization, yet standardized fusion protocols remain limited. While *UAVSwarm* [133] includes up to 23 drones, testing on larger fleets (50+)

remains uncommon, constraining research on scaled co-operative tracking and interception. Moreover, stealth or jamming scenarios—emerging as key topics in anti-UAV forums [89, 162, 164]—are still sparsely represented in open-source datasets. High-resolution streams (e.g., *Det-Fly* [166] and *AOT* [10]) also challenge onboard systems in terms of storage and processing, prompting further research into compression and hardware acceleration [10, 11].

3. UAV Classification Approaches

Table 2 synthesizes representative UAV classification methods covering radio frequency (RF), audio, radar, vision, and LiDAR modalities. These methods range from simple drone-presence detection [87] to more nuanced classifica-

Table 3. Representative UAV Detection Approaches by Modality. *Disclaimer:* All entries summarize our own interpretation of each paper’s methodology and reported outcomes. Since authors use distinct datasets, labels, and metrics, **no direct numerical comparison** is valid. *Abbrev.:* UAV = Unmanned Aerial Vehicle, CNN = Convolutional Neural Network, PSD = Power Spectral Density, LSTM = Long Short-Term Memory, DNN = Deep Neural Network, RNN = Recurrent Neural Network, YOLO = You Only Look Once, FMM = Feature Modeling Module, FCM = Feature Capture Module, LGFFM = Local-Global Feature Focusing Module, EMA = Efficient Multi-scale Attention, DCNv2 = Deformable Convolution v2, CSDE = Cross Stage Detail Enhance, CSP-PMSA = Cross Stage Partial, Partial Multi-scale, DERS = Detail Enhanced Rep Shared, MCJT = Multi-Consistency Joint Tracker, RCMC = Robust Motion Constraint Module, FSRM = Flexible Spatial Remapping Module, ATUS = Adaptive Template Update Strategy, MFCC = Mel Frequency Cepstrum Coefficient, RCNN = Region-based CNN, FPN = Feature Pyramid Network, SWIN = Shifted Window Transformer, DETR = Detection Transformer, OEN = Object Enhancement Net, IR = Infrared, LiDAR = L, F1 = harmonic mean of precision & recall, mAP@0.5 = mean Average Precision at 0.5 IoU, mAP@0.5:0.95 = mean Average Precision at IoU in [0.5:0.95], AP@0.5 = Average Precision at 0.5 IoU, SOTA = state-of-the-art, RA = Radar.

Ref. (Yr.)	Mod.	Method	Dataset	Key Observations (Our Reading)
<i>Radar-Based</i>				
Zhao et al. (2022) [165]	RA	CNN + LSTM	mDrone [165]	Exploits Doppler signals from propellers for real-time detection and 3D tracking at 10 Hz. Achieves ≈ 9 cm localization error, outperforming five classic radar baselines (e.g., 3D MUSIC) by $\approx 66\%$ in accuracy.
<i>Vision-Based</i>				
Seidaliyeva et al. (2020) [110]	RGB	Background subtraction + MobileNetV2	Drone-vs-Bird [29], BirdDetection [154]	Combines Drone-vs-Bird with a bird dataset to enhance UAV-vs-bird discrimination. Reached 0.56 F1 at IoU=0.5 (9 FPS). Performance is improved upon by works using more advanced YOLO variants (e.g., [31]).
Dadboud et al. (2021) [31]	RGB	YOLOv5x (PANet neck, mosaic augment)	Det-Fly [166], Drone-vs-Bird [29]	Combined Drone-vs-Bird + Det-Fly, achieving 0.98 mAP@0.5 with 0.96 recall. Improves upon [110] on Drone-vs-Bird, though different training splits limit exact comparison.
Sangam et al. (2023) [108]	RGB	CSPDarkNet-53 VideoSwin	+ FL-Drones [103], NPS-Drones [68], AOT [10]	On FL-Drones: 0.89 precision, 0.85 recall, 0.87 F1, 0.84 AP@0.5. On NPS-Drones: 0.94 precision, 0.92 recall, 0.93 F1, 0.93 AP@0.5. On AOT: 0.85 precision, 0.76 recall, 0.80 F1, 0.82 AP@0.5. Comparable to Rebbapragada et al. [98] under similar setups.
Tu et al. (2024) [128]	RGB	YOLOv8 + FCM + FMM	DUT Anti-UAV [163]	AP@0.5=0.913, AP@0.5:0.95=0.612, below the best YOLO-based methods on this dataset (e.g., [53]) but highlights a lightweight, efficient design.
Cheng et al. (2024) [26]	RGB	YOLOv5s + LGFFM	DUT Anti-UAV [163]	mAP@0.5=0.943. Improves the recall rate of 2.96% and the precision of 1.41% compared with YOLOv5s. Slightly outperforms [128] on the same benchmark. Low-complexity approach with 4.4 ms/frame inference.
Bo et al. (2024) [16]	RGB	YOLOv7-tiny + Inception-NeXt + SPPFCSPC-SR + “Get-and-Send”	DUT Anti-UAV [163]	mAP@0.5=0.932 at 104 FPS. Balances fast inference and high accuracy, though slightly lower mAP@0.5 than [26] and [53].
Elsayed et al. (2024) [38]	RGB	YOLOv6 + LERFNet + large-kernel LAM	DUT Anti-UAV [163]	mAP@0.5=0.936, with 0.953 precision. Surpasses [16] in accuracy, albeit at presumably higher resource use than the YOLOv7-tiny approach.
Huang et al. (2024) [53]	RGB	YOLOv8 + ghost conv + EMA + DCNv2	Det-Fly [166], DUT Anti-UAV [163]	On DUT Anti-UAV, yields 0.971 mAP@0.5, highest among YOLO-based reports here ([16, 26, 38, 128]). On Det-Fly, obtains 0.934 mAP@0.5.
Sun et al. (2024) [118]	IR	YOLOv8 w/ triple-input, BiFormer attention, small-object layer	Hui [55], Anti-UAV [58], Anti-UAV410 [51]	On Hui dataset: 0.943 mAP@0.5 at 72 FPS. Generalizes well to Anti-UAV (0.948 mAP@0.5) and Anti-UAV410 (0.927 mAP@0.5). Demonstrates strong IR-based detection across multiple IR benchmarks.
Wisniewski et al. (2024) [141]	RGB	Faster R-CNN	MAV-VID [100], Anti-UAV [58], Drone-vs-Bird [29]	AP@0.5=0.970 on MAV-VID, AP@0.5=0.498 on Drone-vs-Bird, AP@0.5=0.678 on Anti-UAV.
Zhou et al. (2024) [169]	RGB	YOLOv8 + multi-scale spatial attention + HDC	Det-Fly [166]	Achieved 0.815 precision, 0.849 recall. Gains of +0.05 precision/+0.06 recall relative to YOLOv8 baseline. Lags behind [53] and [46] in absolute mAP on Det-Fly, but emphasizes robust small-object detection.
Rebbapragada et al. (2024) [98]	RGB	SWIN + DETR + OEN	FL-Drones [103], NPS-Drones [68], AOT [10]	Matches Sangam et al. [108] in precision/recall/AP@0.5 = 0.89/0.85/0.84 on FL-Drones and 0.94/0.92/0.93 on NPS-Drones with a transformer-based pipeline. On AOT dataset, obtains 0.85 precision, 0.76 recall, AP@0.5=0.82 comparable to [108].
Hao et al. (2025) [46]	RGB	YOLOv8 w/ CSDE, CSP-PMSA, OmniKernel, DERS head	Det-Fly [166], MOT-Fly [28]	On MOT-Fly, hits mAP@0.5=0.949 with only 1.13M parameters. On Det-Fly, achieves 0.964 mAP@0.5—slightly exceeding [53] at 0.934, though using different training splits.
<i>Multi-Modal</i>				
Svanström et al. (2021) [122]	RGB, IR, Aud.	YOLOv2 (visible) + MFCC (audio) + LSTM	Halmstad Drone [122]	Achieved F1 $\approx$ 0.61–0.88 across different distance with thermal IR sensor, F1 $\approx$ 0.69–0.86 with visible camera, F1 $\approx$ 0.93 with audio detector. Fusion cut false alarms vs. single-sensor methods, illustrating multi-modal benefits.
Wu et al. (2024) [143]	RGB, RA	Attention-based Spatiotemporal Fusion Net	None public	Achieved 93.0% precision, 83.7% recall, 88.7% mAP in occluded scenes at 20 FPS. Outperforms vision-only detectors by fusing radar signals, though dataset not publicly released.

tions of UAV types, flight modes, and orientations [109]. Although comparison is difficult due to inconsistent experimental protocols, modality-specific considerations—such as spectral noise or lighting variability—motivate distinct architectural and data-processing strategies. Below, we highlight key insights and emerging trends.

(1) RF-Based Classification. RF-focused approaches exploit the electromagnetic signatures emitted by UAVs to detect, identify, and even recognize specific flight modes. Al-Sa’d et al. [3] introduced the DroneRF dataset, demonstrat-

ing high accuracy (99.7%) on a 2-class task but lower performance (46.8%) on a more granular 10-class task. Subsequent work by Al-Emadi et al. [1] confirmed CNNs’ viability for DroneRF, achieving 99.8% (2-class) and 59.2% (10-class). Recent studies have explored advanced feature engineering or ensemble learning: for instance, Nemer et al. [86] outperformed CNN-based baselines via hierarchical ensembles, while Li et al. [72] incorporated bispectrum representations with contrastive learning. Inani et al. [56] achieved near-perfect 10-class accuracy (99.51%) by fus-

ing PSD features, a 1D-CNN, and XGBoost. Despite these advancements, real-world deployments face challenges including spectral congestion, adversarial jamming, and domain shifts [9, 37, 132].

(2) Audio-Based Classification. Audio pipelines distinguish UAVs based on rotor noise and other acoustic signatures. Al-Emadi et al. [2] introduced the DroneAudio dataset, reporting 96.38% (2-class) and 92.94% (3-class) accuracy using CNNs. Katta et al. [61] subsequently demonstrated the benefits of LSTM, CRNN, and Transformers, with LSTM attaining 98.93% (2-class). Meanwhile, Xiao et al. [148] employed a temporal CNN+RNN to handle MMAUD's audio modality, reaching 98.0% accuracy alongside strong pose estimation results. Noise from wind, background chatter, or multiple UAVs remains an obstacle in practical scenarios [14, 57], motivating research into denoising filters and multi-microphone arrays [119].

(3) Radar-Based Classification. Radar offers micro-Doppler signatures that persist under reduced visibility or nighttime conditions. Early studies (e.g., Mendis et al. [81]) validated radar-based UAV classification using deep belief networks, though limited dataset availability inhibited broad adoption. CNN- and LSTM-based pipelines now dominate, focusing on micro-Doppler representations to improve robustness in cluttered environments [132, 166]. Nonetheless, radar hardware cost and calibration constraints remain practical challenges [132].

(4) Vision-Based Classification. Computer vision pipelines target UAV recognition via optical images, often focusing on manufacturer types (e.g., DJI Phantom, Mavic). Wisniewski et al. [139] achieved 92.4% accuracy on three drone types (Phantom, Mavic, Inspire) and illustrated synthetic imagery's viability [140]. Other studies address issues of small object size and motion blur, incorporating advanced detectors or domain adaptation [109, 130]. While performance in controlled scenarios is high, vision-based methods degrade under extreme illumination, adverse weather, or dense clutter [18, 33].

(5) Multi-Modal Recognition. Integrating multiple sensors (e.g., RF+Audio [43], Radar+LiDAR+Vision [156], or Audio+Vision [147]) can mitigate single-sensor blind spots. Deng et al. [32] fused fisheye RGB, radar, and LiDAR data from MMAUD, reaching 81.36% accuracy on a 4-class task, while Xiao et al. [147] combined audio and RGB streams to achieve 99.3% accuracy and sub-meter pose errors. Although multi-modal setups improve robustness, they also require careful calibration, synchronization, and greater computational resources [132, 156].

Key Insights & Challenges. Across these approaches [18, 33, 130, 132, 166], each modality faces inherent constraints: (1) *RF-based solutions* excel at flight-mode or type identification [3, 9], but radio congestion and potential adversarial attacks are critical hurdles [37, 132]. (2) *Audio-based meth-*

ods achieve high accuracy in controlled settings yet suffer from significant performance drops in high-noise or windy environments [2, 14, 57]. (3) *Radar-based pipelines* bypass visual occlusions but incur higher costs and calibration demands [81, 132]. (4) *Vision-based approaches* generally provide fine-grained recognition but degrade with lighting or weather changes [130, 139]. (5) *Multi-modal fusion* can mitigate single-sensor weaknesses and reduce false positives [32, 43] yet raises complexity in real-time deployment [156]. Additionally, adversarial robustness remains an open problem, particularly where sensors or neural networks may be fooled by spoofing or deceptive signals [61, 80]. A continuing research direction involves efficient, low-latency architectures [132, 148] that can handle on-board or edge-based processing for rapid UAV detection and classification.

4. UAV Detection Approaches

Table 3 summarizes UAV detection pipelines spanning radar, vision, and multi-modal solutions. Each modality addresses distinct obstacles—such as small object size, cluttered environments, and adverse weather—offering unique advantages and trade-offs. Although direct numerical comparisons are challenging due to varying datasets and evaluation protocols, these methods collectively illustrate emerging trends in detection algorithms and system design. We discuss salient insights from the literature below.

(1) Radar-Based Detection. Radar methods detect UAVs by exploiting micro-Doppler signatures, providing resilience under low-visibility or harsh weather conditions [132, 165]. For instance, Zhao et al. [165] achieved decimeter-level localization at 10 Hz by combining CNNs with LSTMs, outperforming conventional radar baselines. However, radar sensors can suffer from interference and clutter, particularly in urban areas or heavily congested electromagnetic environments [143].

(2) Vision-Based Detection. Vision-based techniques remain widely adopted, benefiting from readily available cameras and sophisticated CNN-based detectors [124, 170]. Early work centered on region-based methods [99] and YOLO architectures [17, 135], gradually evolving into more lightweight or attention-oriented variants [26, 46, 53]. While mosaic augmentation and specialized modules can improve small-object detection [26, 31], vision-based approaches remain vulnerable to occlusions, clutter, and extreme distances. Researchers have thus explored thermal and infrared (IR) imaging [85, 118] or speed-accuracy trade-offs for real-time embedded scenarios [16, 38].

(3) Multi-Modal Fusion. Single-modality approaches often fail under suboptimal conditions—low-visibility (vision), interference (radar), or noise (audio) [2, 129]. Consequently, multi-modal fusion has gained traction as a means of leveraging complementary sensor data [12, 120]. Svanström et al. [120] showed that augmenting YOLO-

Table 4. Representative UAV Tracking Approaches. *Disclaimer:* All entries summarize our own interpretation of each paper’s methodology and reported outcomes. Since authors use distinct datasets, labels, and metrics, **no direct numerical comparison** is valid. *Abbrev.:* UAV = Unmanned Aerial Vehicle, CNN = Convolutional Neural Network, RNN = Recurrent Neural Network, SVM = Support Vector Machine, DNN = Deep Neural Network, LSTM = Long Short-Term Memory, GMM = Gaussian Mixture Model, IR = Infrared, TIR = Thermal Infrared, ID = Identification, Radar = RA, LiDAR = L, SOTA = State of the Art, YOLO = You Only Look Once, YOLOX = You Only Look Once Extended, DFSC = Dual-Flow Semantic Consistency, LoFTR = Detector-Free Local Feature Matching with Transformers, SiamRPN++ = Siamese Region Proposal Network++, SiamSTA = Siamese Spatio-Temporal Attention tracker, SiamSRT = Siamese Search Region-free Tracker, SiamDT = Siamese Dual-semantic Tracker, SiamFusion = Siamese Modality-Fusion Tracker, DCF = Discriminative Correlation Filter, CDCF = Change Detection-based Correlation Filter, CFRM = Channel Feature Refinement Module, BYTE = Bounding-box Embedding Tracker with Association, STG-RPN = Spatio-Temporal Guided Region Proposal Network, ID-RCNN = Instance Discrimination Region-based CNN, FPS = Frames Per Second, Acc = Accuracy, F1 = F1 Score, TLD = Tracking-Learning-Detection, DFSC = Dual-Flow Semantic Consistency, ATUS = Adaptive Template Updating Strategy, EECF = Eagle-Eye Correlation Filter, ECA = Efficient Channel Attention, STFTrack = Spatio-Temporal Focused Tracker, CAM = Channel Attention Module, MCJT = Multi-Consistency Joint Tracker, α -IoU = Alpha Intersection over Union, Swin = Shifted Window Transformer, MSE = Mean Squared Error, SPLT = Self-paced Learning Tracker.

Ref. (Yr.)	Mod.	Method	Dataset	Key Observations (Our Reading)
<i>Detection-Based Trackers</i>				
Jiang et al. (2023) [58]	RGB, IR	GlobalTrack + DFSC	Anti-UAV [58]	Released 318-sequence RGB-T UAV tracking benchmark. Achieved the best state accuracy score of 74.33% on the validation set and 65.41% on the test set
Wu et al. (2024) [144]	RGB	DCF + dual-fovea + contrast + opponent-color (EECF)	Anti-UAV [58], USC-Drone [24]	Real-time EECF tracker (86 FPS) via dual-fovea design. Achieved the state accuracy score of 66.42% on Anti-UAV, 70.45% on USC-Drone.
Wang et al. (2024) [134]	RGB	YOLOX + Swin-T + α -IoU + BYTE	UAVSwarm [133]	Detection-driven approach with α -IoU + ByteTrack. Achieved 78.2% tracking accuracy on UAVSwarm test split.
Deng et al.(2024) [32]	RGB, RA, L	YOLOv9 + EfficientNet-B7 + LSTM	MMAUD [156]	Integrated YOLOv9 + EfficientNet-B7 + LSTM/Kalman for classification and tracking. Achieved 2.21 Pose MSE Loss on 3D tracking of MMAUD and ranked 1st in UG2+ Challenge.
<i>Siamese (Contrastive) Trackers</i>				
Huang et al. (2021) [49]	IR	SiamSTA	1st / 2nd Anti-UAV Challenges [58] [162]	Achieved 74.46 and 67.30 scores on 1st / 2nd Anti-UAV test-dev sets, ranking 1st. Extended SiamSTA with spatio-temporal attention + motion estimation.
Li et al. (2022) [71]	IR	SuperDiMP (ResNet50)	2nd Anti-UAV Challenge [162]	Used SuperDiMP with dual-branch IR fusion. Achieved +5.84% SR and +4.52% PR over baseline. Focused on contrast inversion robustness.
Chen et al. (2022) [22]	IR	SiamSTA + CDCF	1st Anti-UAV Challenge [58]	SiamSTA with CDCF and 3-stage redetection. Achieved top precision in 2nd Anti-UAV Challenge. Achieved 68.26 and 76.11 scores on Anti-UAV test and validation sets. Robust on fast/small UAVs.
Zhang et al. (2023) [161]	RGB, IR	SiamFusion	Anti-UAV [58]	Achieved overall 63.37 score on Anti-UAV with 14 FPS. Developed a dual-fusion tracker with modality + decision-level fusion. Used Swin backbone + Local-Global Converter.
Xie et al. (2023) [149]	IR	FPN-based Siamese + STG-RPN + ID-RCNN (STF-Track)	Anti-UAV [58]	STFTrack using STG-RPN and ID-RCNN. Achieved 91.2% precision, 66.6% success @ 12.4 FPS. Addressed distractors and thermal crossover.
Zhang et al. (2024) [158]	IR	OTrack + MCJT (RMCM + FSRM + ATUS)	1st / 2nd / 3rd Anti-UAV Challenges [58] [162] [164]	Achieved 69.65% Acc, 80.45% Acc, 65.88% Acc and 50.28% Acc on the 1st Anti-UAV test set, 1st Anti-UAV test-dev, 2nd Anti-UAV test-dev, and 3rd Anti-UAV validation set. Proposed MCJT (RMCM + FSRM + ATUS) with adaptive template and multi-consistency. Ran at 44 FPS.
Huang et al. (2024) [50]	IR	SiamSRT + C-C RPN + S-L RCNN + TMB + SCFD	Anti-UAV [58]	Achieved 71.64 and 80.04 scores on Anti-UAV testing and validation sets. Global Siamese tracker using Swin-T + consistency modules (TMB + SCFD). Avoided template degradation; outperformed multiple SOTA trackers.
Huang et al. (2024) [51]	IR	SiamDT (Swin Transformer + Dual-Semantic RPN + Versatile R-CNN)	Anti-UAV410 [51]	Introduced Anti-UAV410 dataset. Proposed SiamDT with dual semantic reasoning + background suppression. Achieved 68.19 and 71.65 scores on its testing and validation sets.
<i>Hybrid (Detection + Siamese) Trackers</i>				
Cheng et al. (2022) [25]	IR	Multi-Hybrid Attention + SiamRPN++ + YOLOv5 (SiamAD)	Anti-UAV [58]	SiamAD with CSAM/CAM + hierarchical discriminator + YOLOv5. 88.4% DP @ 37.1 FPS. Improved long-term IR tracking.
Yu et al. (2023) [155]	IR	Transformer + YOLOv5 + LoFTR + GMM + OTrack	1st / 2nd Anti-UAV Challenges [58] [162]	Achieved 2nd place in Anti-UAV Challenge (3rd) with 68.8 score. And 75.13% and 80.38% accuracy on the 1st and 2nd test dev, individually. Integrated OTrack, YOLOv5, LoFTR, GMM for multi-stage IR tracking.

based RGB or thermal imaging with audio features increased F1 scores and reduced false alarms. Similarly, Wu et al. [143] fused radar and RGB streams via attention-based spatiotemporal networks to improve occlusion handling. Practical adoption, however, demands meticulous sensor synchronization, balanced hardware costs, robust computational efficiency, and real-time data association [98].

Key Insights & Challenges. From prior work [124, 132, 170], several recurring themes emerge. 1) *Small UAV Detection:* Limited pixel coverage and subtle micro-Doppler signatures complicate detection [20, 47, 77], prompting multi-scale feature extraction, deformable convolutions, and transformer backbones [26, 53, 124]. Ensuring real-time operation at high accuracy remains challenging [132,

135]. 2) *Environmental Factors:* Ambient noise (audio), lighting (vision), and radio interference (radar) degrade performance [2, 113, 129, 165], highlighting the need for advanced clutter suppression and robust sensor hardware. 3) *Generalization vs. Specialization:* YOLO-based models offer generalizable frameworks [26, 31, 46], but specialized domains (e.g., stealth UAVs) may require niche architectures or domain-adaptive strategies [16, 143]. 4) *Multi-Modal Fusion:* Complementary sensors effectively reduce false positives and enhance accuracy [120, 143], although issues like sensor alignment, real-time latency, and hardware cost must be resolved [12, 98]. 5) *Benchmarking & Deployment:* Heterogeneous data sources, labeling formats, and test conditions [35, 93, 170] complicate standard-

ized evaluations. Though laboratory demonstrations show promise, real-time performance in crowded outdoor settings remains largely underexplored [39, 42]. Overcoming these challenges will be crucial for field-ready anti-UAV systems.

5. Anti-UAV Tracking Approaches

Table 4 reviews prominent UAV tracking methods spanning detection-based, Siamese-based, and hybrid trackers, employing RGB, IR, and multi-modal inputs. Although direct numerical comparisons are hampered by dataset and metric variations, these approaches collectively represent the state of the art in UAV tracking.

(1) Detection-Based Trackers. Detection-based pipelines usually prioritize real-time performance and utilize efficient object detectors. For example, Wu et al. [144] proposed EECF, a dual-fovea correlation filter tracker operating at 86 FPS with a state accuracy score of 66.42% on Anti-UAV [58] and 70.45% on USC-Drone [24]. Wang et al. [134] integrated YOLOX [44] with Swin Transformer [77] and ByteTrack, achieving 78.2% accuracy on UAVSwarm [133]. Deng et al. [32] combined YOLOv9 [136] and EfficientNet-B7 [123] with LSTM/Kalman filtering for 3D tracking, earning top-ranked pose MSE on MMAUD [156] and winning the UG2+ Challenge [90].

(2) Siamese (Contrastive) Trackers. Siamese trackers excel in learning robust feature similarities and handling fast or evasive UAV motion. Huang et al. [49] claimed first-place in the 1st and 2nd Anti-UAV Challenges [58, 162] with SiamSTA, which incorporates sophisticated spatio-temporal attention and motion estimation. Li et al. [71] augmented SuperDiMP with a dual-branch IR fusion strategy, boosting precision by 4.52% in the 2nd Anti-UAV Challenge. Chen et al. [22] further enhanced SiamSTA with CDCF and a multi-stage re-detection mechanism, excelling against small, fast UAVs. Beyond single-modality, Zhang et al. [161]’s SiamFusion seamlessly integrated RGB and IR streams for a 63.37 score on Anti-UAV, illustrating the tangible value of complementary sensor modalities.

(3) Hybrid (Detection + Siamese) Trackers. Hybrid methods combine detection pipelines with Siamese matching, capitalizing on each paradigm’s respective strengths. Cheng et al. [25] introduced SiamAD, a synthesis of YOLOv5, multi-hybrid attention modules, and SiamRPN++, achieving 88.4% detection precision at 37.1 FPS on the Anti-UAV dataset [58]. Yu et al. [155] integrated Transformers, YOLOv5, LoFTR, GMM, and OTrack for second place in the 3rd Anti-UAV Challenge [164], affirming the practical effectiveness of multi-stage IR-based tracking.

Key Insights & Challenges. Although real-time Siamese trackers [111] and detection-based pipelines [40] can sustain high frame rates, they often struggle under clutter, occlusion, or swarm conditions, thereby motivating more adaptive architectures [23, 150]. Stealth maneuvers and

jamming further complicate UAV tracking, emphasizing the necessity for robust domain adaptation and dynamic cue prioritization [34]. While RGBT sensor fusion [67, 125] mitigates individual sensor shortcomings, incorporating LiDAR or radar [32] inevitably increases calibration overhead and computational load. Existing benchmarks rarely capture large-scale swarms, diverse weather conditions, or severe occlusions, consequently restricting methodical progress [145]. Transformer-based solutions [23, 30, 146] address complex motion patterns and partial occlusion but can lag in real-time performance, necessitating specialized hardware optimizations for on-board or edge devices. Extending multi-object trackers (e.g., ByteTrack [159]) to larger UAV swarms remains challenging due to association uncertainty and domain-specific constraints. Ultimately, next-generation trackers will demand faster inference, improved sensor fusion, and broader datasets to achieve reliable anti-UAV performance in real-world conditions.

6. Emerging Techniques & Future Directions

Recent UAV research has embraced advanced machine learning paradigms to boost performance, scalability, and autonomy. Figure 2 categorizes these innovations into five key themes: (i) diffusion models for data synthesis and domain adaptation, (ii) vision-language models for semantic understanding, (iii) self-/unsupervised learning, (iv) multi-modal fusion and sensor integration, and (v) reinforcement learning for autonomous control. We outline critical developments and future opportunities below.

Diffusion Models for Data Synthesis & Domain Adaptation. Diffusion models [48, 116] generate high-fidelity UAV imagery to address data scarcity and enhance domain diversity, including text-conditioned approaches [97, 105]. Recent advances in faster sampling [95, 106] support near real-time augmentation, while diffusion-based domain adaptation [13, 94, 160] narrows the gap between synthetic and real-world data. Future research may focus on on-the-fly adaptation from live sensor streams, enabling continuous refinement under dynamic UAV operating conditions.

Vision-Language Models & Semantic Context. Vision-language (VL) methods integrate textual cues with visual features to facilitate open-vocabulary UAV detection and classification [45, 96, 157, 167]. UAV-specific adaptations [52, 73, 76] incorporate textual descriptions on flight profiles, mission goals, or payload attributes, thereby *significantly* enriching threat assessment. By leveraging semantic narratives (e.g., infiltration routes, suspect cargo), future VL-based systems can *swiftly* adapt to novel UAV categories and enhance situational awareness.

Self-Supervised & Unsupervised Learning. Self- and unsupervised approaches alleviate dependence on labeled aerial data across varying altitudes, sensors, and weather [64, 65, 152]. Recent work addresses swarm-wide gener-

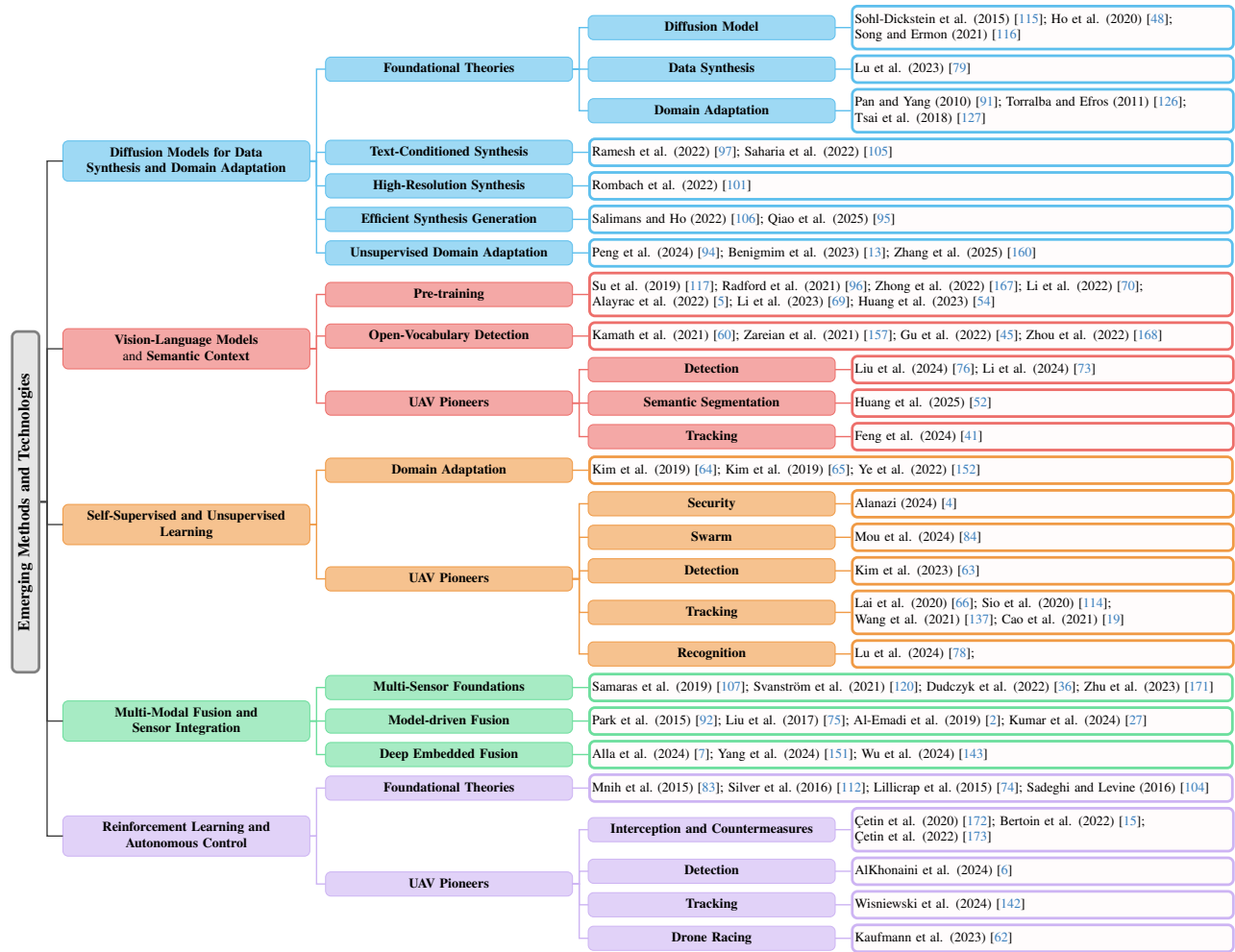


Figure 2. Taxonomy for emerging methods and technologies in UAV research.

alization [84], *further* strengthening multi-UAV detection [63], tracking [66, 137], and recognition [114], as well as security [4]. Moving forward, large-scale self-supervised pre-training that *rapidly* adapts to domain shifts in real time may enable *ongoing* responses to evolving UAV behaviors and stealth tactics.

Multi-Modal Fusion & Sensor Integration. Combining sensors (e.g., RGB, IR, radar, and audio) *significantly* boosts UAV detection and tracking robustness [82, 92, 107, 171]. Classical fusion [75, 92] is now augmented by deep embedding methods [7, 151], allowing adaptive sensor weighting under real-time constraints. However, *maintaining* precise synchronization [143] and carefully balancing power or bandwidth remain essential considerations for *robust*, large-scale deployments.

Reinforcement Learning & Autonomous Control. Reinforcement Learning (RL) has proven effective for adaptive UAV decision-making in dynamic environments, building on foundational breakthroughs [83, 112]. Modern UAV-centric RL research tackles interception [15, 173], detection

[6], tracking [142], and high-speed racing [62], frequently integrating generative or multi-modal inputs for rapid policy updates. Future directions may emphasize hierarchical or multi-agent RL for large-scale swarm control, balancing cooperative behavior with adversarial countermeasures and mission-specific constraints.

7. Conclusion

This survey examined the evolving anti-UAV landscape, focusing on detection, tracking, and recognition. We highlighted recent progress in sensor fusion, deep learning, and multi-modal systems, emphasizing the urgency of robust, real-time solutions under adverse conditions. Analysis of representative datasets revealed gaps in adversarial robustness and scalable swarm tracking. Key insights underscore cross-modal integration, resilience to adversarial tactics, and the promise of self-supervised and reinforcement learning. Moving forward, generative modeling, semantic understanding, and adaptive multi-modal frameworks are poised to drive the next wave of anti-UAV innovations.

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